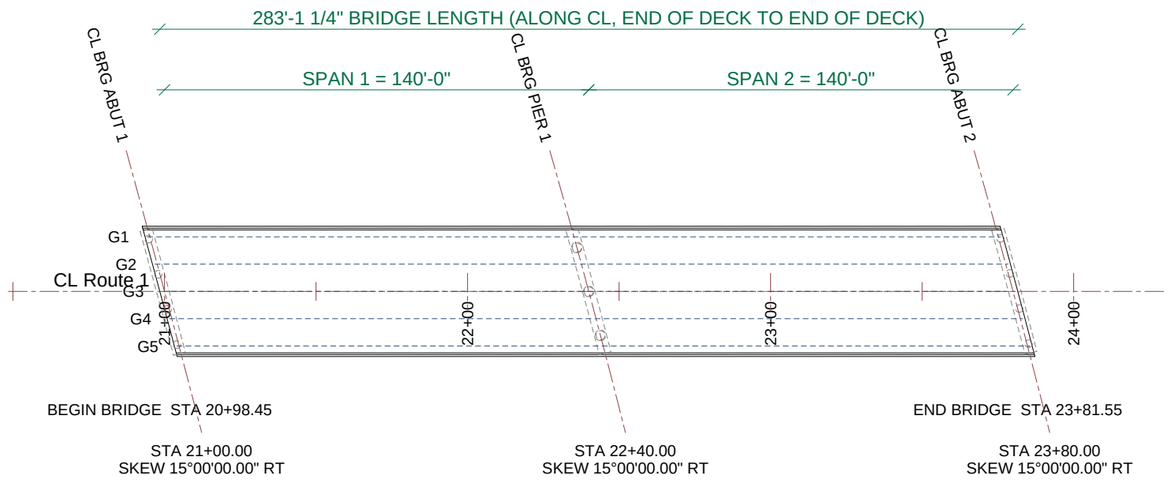
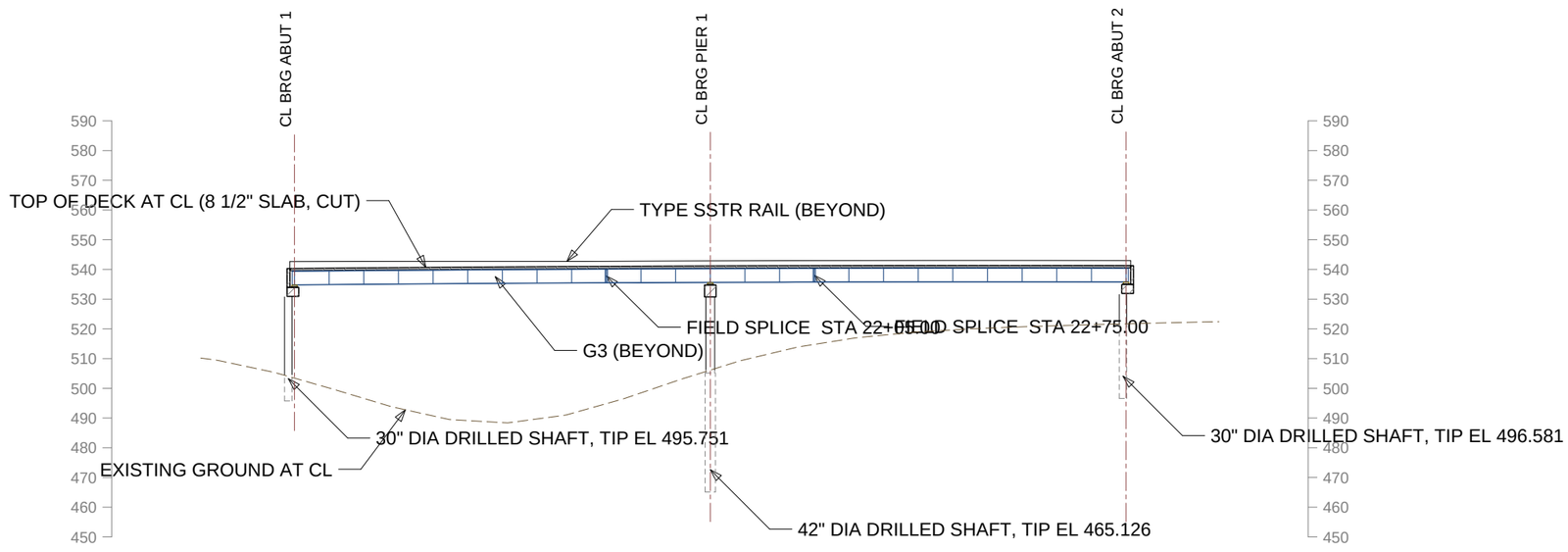
PLAN (DEVELOPED ALONG CL ALIGNMENT)

Stations align with the elevation below



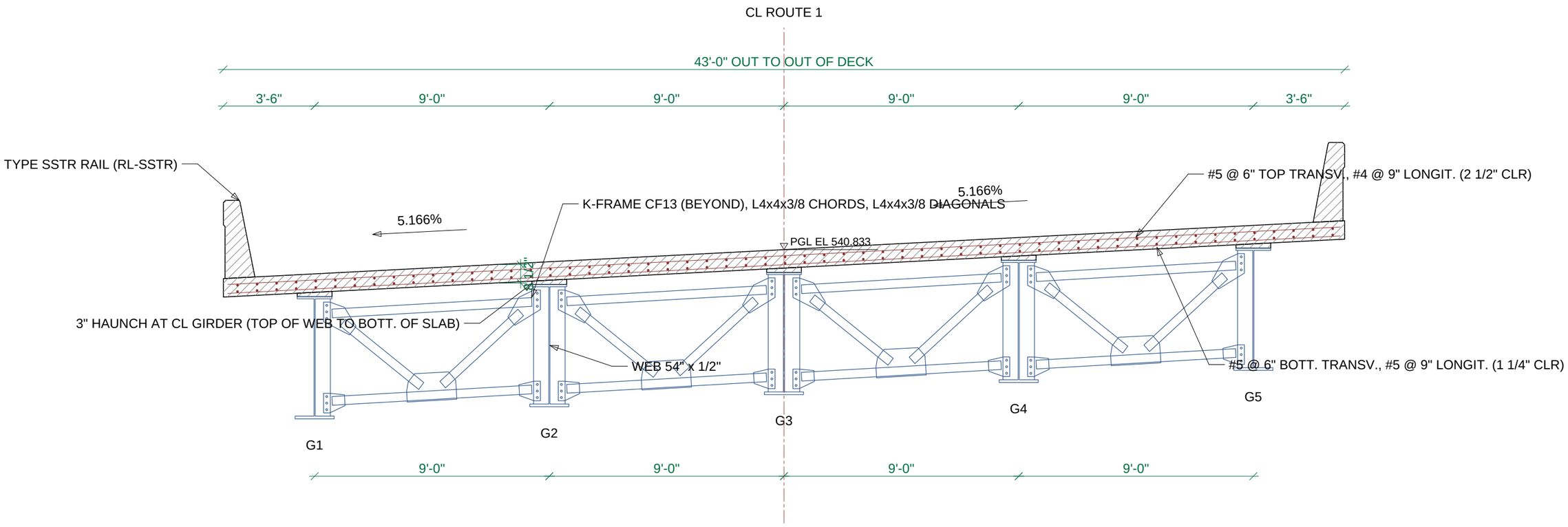


ELEVATION — SECTION ALONG CL ALIGNMENT (LOOKING LEFT)

Scale 1" = 50'-0"

SCALES Plan and elevation: 1" = 50'-0"	PROJECT SAMPLE PROJECT STEEL BRIDGE MODELER	BRIDGE LAYOUT Plan and elevation along the CL alignment SAMPLE BRIDGE	SHEET S-1 (1 OF 14) 10/08/2026

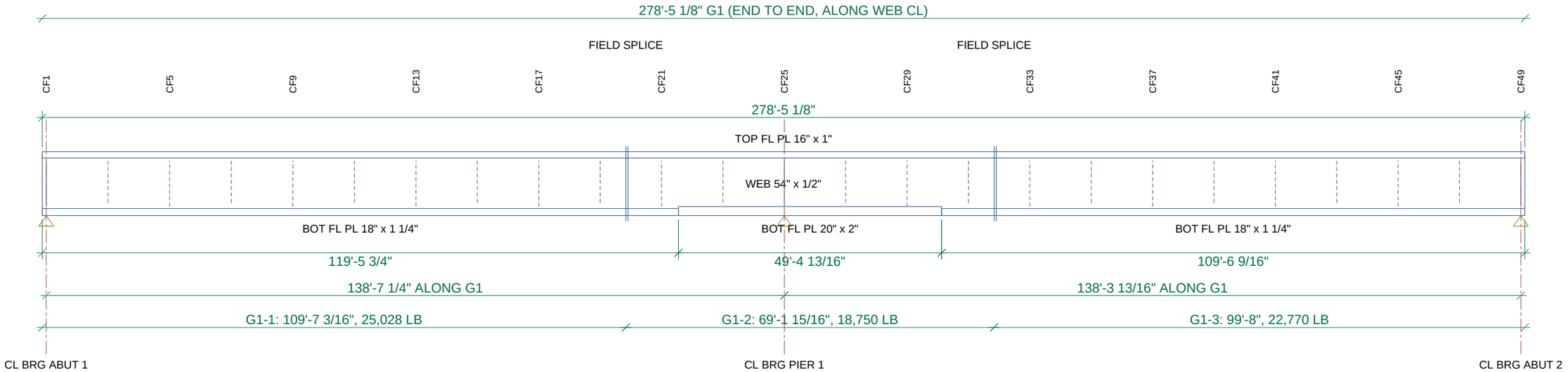
GEOMETRY FROM THE PARAMETRIC MODEL. NOT FOR CONSTRUCTION UNLESS SEALED.



TYPICAL SECTION AT STA 21+70.00

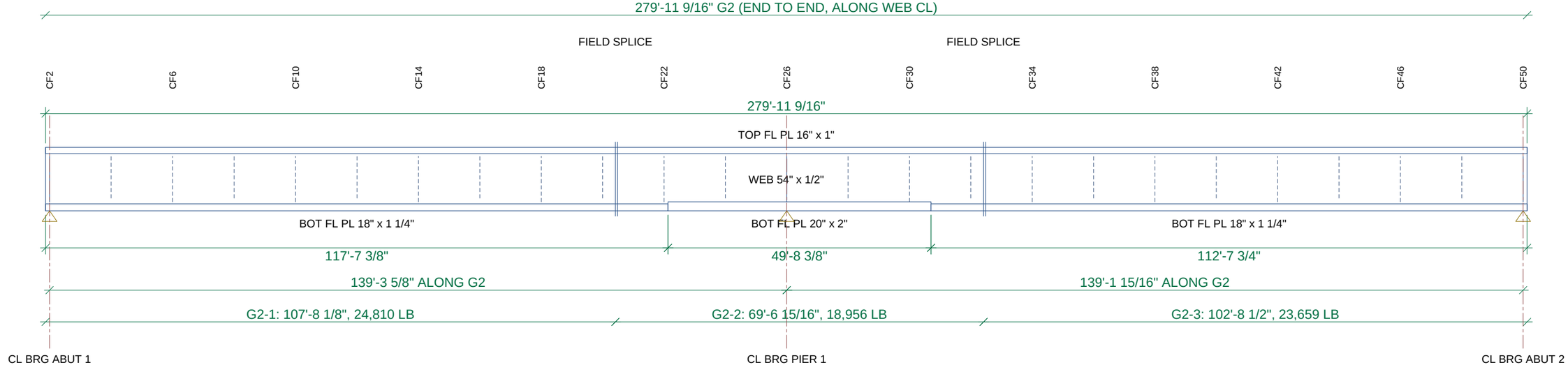
Looking up-station · Scale 1/4" = 1'-0"

SCALES Typical section: 1/4" = 1'-0"	PROJECT SAMPLE PROJECT	TYPICAL SECTION Superstructure SAMPLE BRIDGE	SHEET S-2 (2 OF 14)
GEOMETRY FROM THE PARAMETRIC MODEL. NOT FOR CONSTRUCTION UNLESS SEALED.	STEEL BRIDGE MODELER		10/08/2026



G1 ELEVATION

Developed along the web CL; depth not to scale. Plate changes at radial lines; stiffeners: solid = bearing, dashed = connection / intermediate.



G2 ELEVATION

Developed along the web CL; depth not to scale. Plate changes at radial lines; stiffeners: solid = bearing, dashed = connection / intermediate.

SCALES

G1 elevation: 1" = 20'-0"
G2 elevation: 1" = 20'-0"

GEOMETRY FROM THE PARAMETRIC MODEL. NOT FOR CONSTRUCTION UNLESS SEALED.

PROJECT

SAMPLE PROJECT

STEEL BRIDGE MODELER

GIRDER ELEVATIONS G1 AND G2

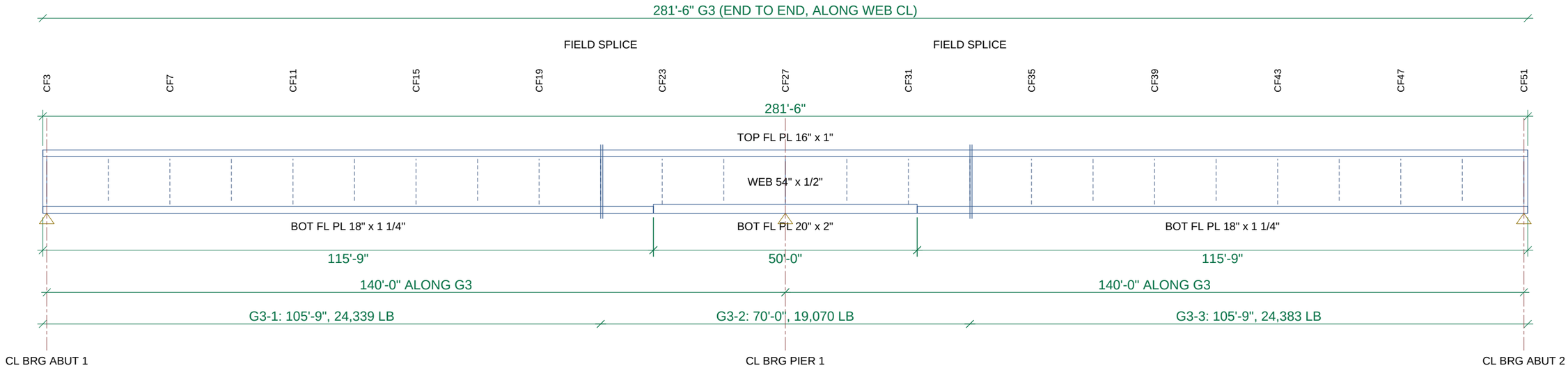
Plate sizes, splices, stiffeners and cross-frame locations

SAMPLE BRIDGE

SHEET

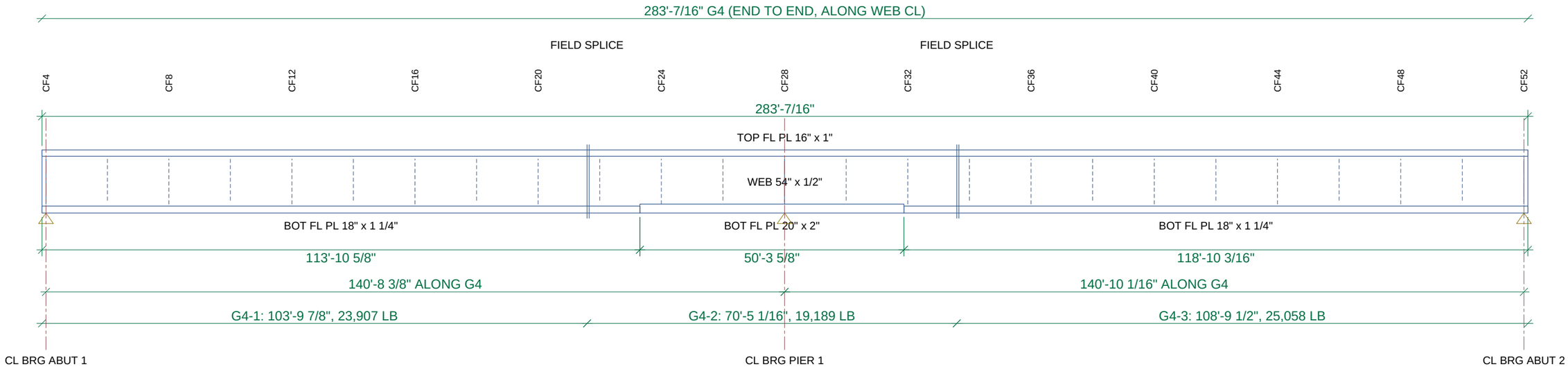
S-3 (3 OF 14)

10/08/2026



G3 ELEVATION

Developed along the web CL; depth not to scale. Plate changes at radial lines; stiffeners: solid = bearing, dashed = connection / intermediate.



G4 ELEVATION

Developed along the web CL; depth not to scale. Plate changes at radial lines; stiffeners: solid = bearing, dashed = connection / intermediate.

SCALES

G3 elevation: 1" = 20'-0"
G4 elevation: 1" = 20'-0"

PROJECT

SAMPLE PROJECT

STEEL BRIDGE MODELER

GIRDER ELEVATIONS G3 AND G4

Plate sizes, splices, stiffeners and cross-frame locations

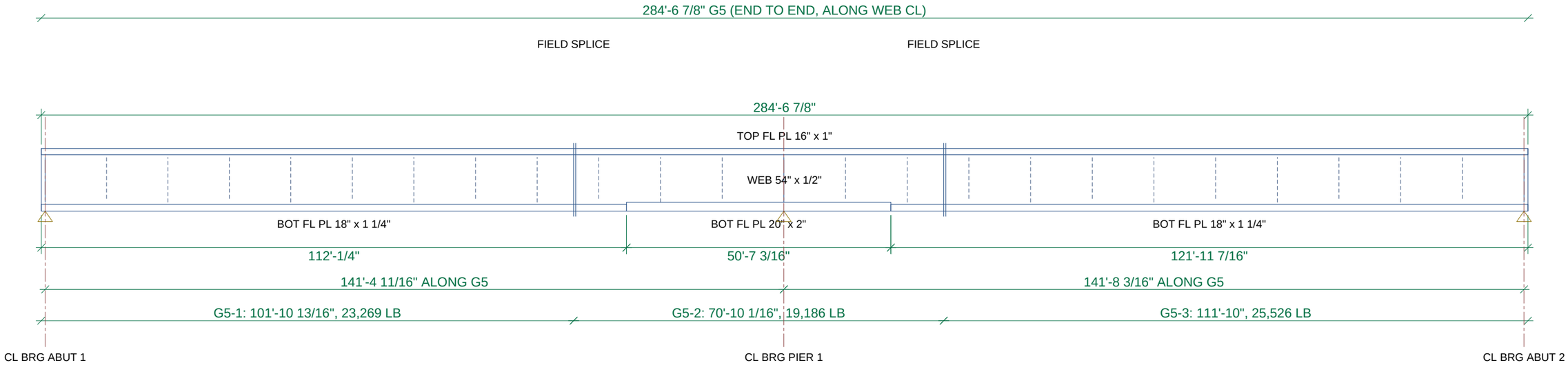
SAMPLE BRIDGE

SHEET

S-4 (4 OF 14)

10/08/2026

GEOMETRY FROM THE PARAMETRIC MODEL. NOT FOR CONSTRUCTION UNLESS SEALED.



G5 ELEVATION

Developed along the web CL; depth not to scale. Plate changes at radial lines; stiffeners: solid = bearing, dashed = connection / intermediate.

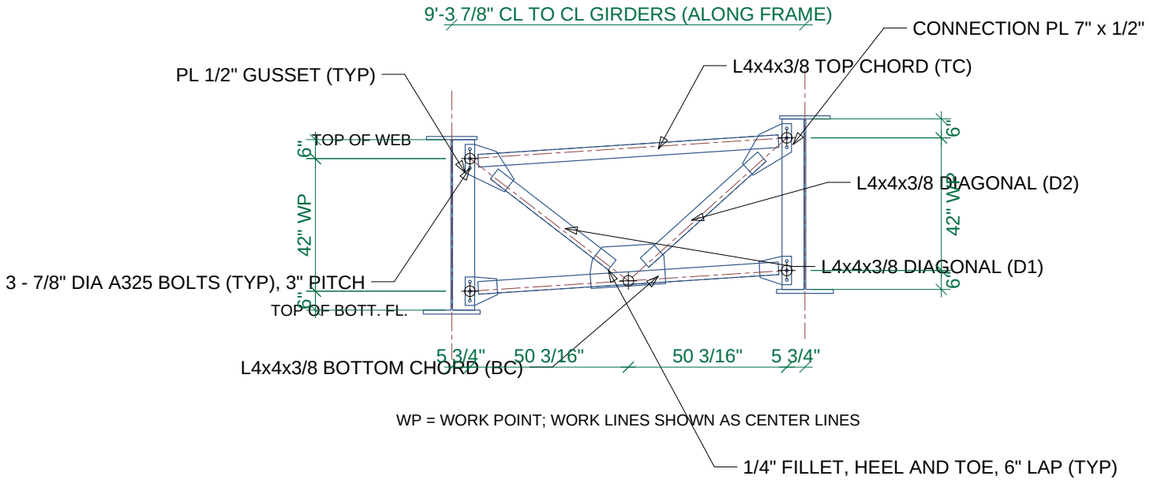
SCALES
G5 elevation: 1" = 20'-0"

PROJECT
SAMPLE PROJECT
STEEL BRIDGE MODELER

GIRDER ELEVATIONS G5
Plate sizes, splices, stiffeners and cross-frame locations
SAMPLE BRIDGE

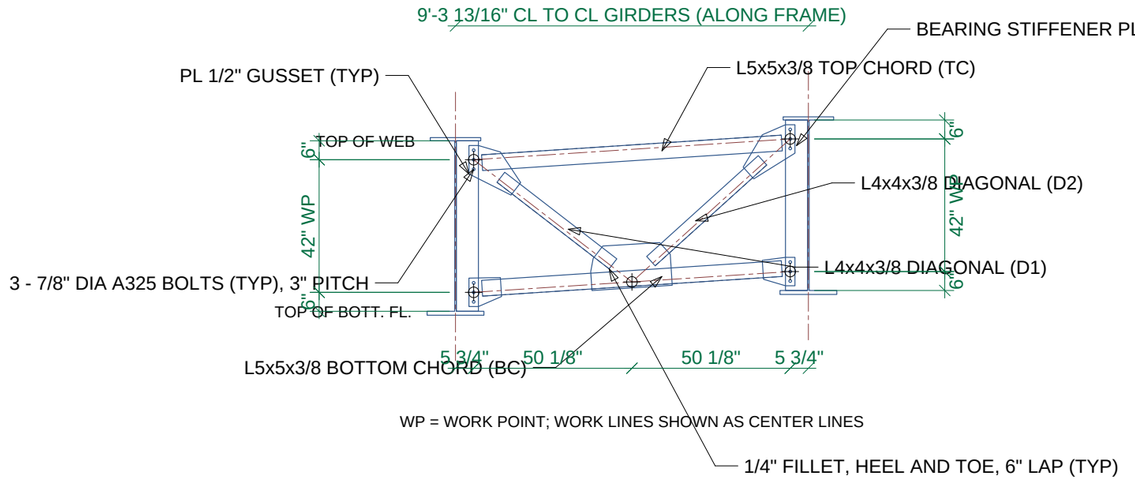
SHEET
S-5 (5 OF 14)
10/08/2026

GEOMETRY FROM THE PARAMETRIC MODEL. NOT FOR CONSTRUCTION UNLESS SEALED.



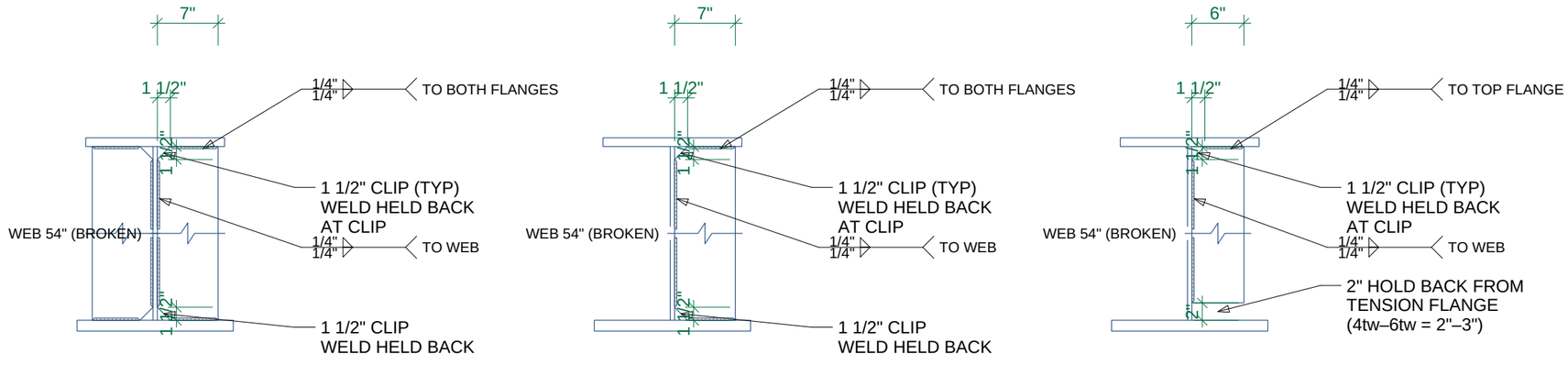
INTERMEDIATE CROSS FRAME — K-FRAME (CF29 SHOWN)

Normal to the flanges; dimensions in the plane of the frame



SUPPORT CROSS FRAME — K-FRAME (CF27 SHOWN)

Plumb, in the plane of the support line



BEARING STIFFENER

PL 7" x 7/8" (PAIR)
MILL TO BEAR ON BOTTOM FLANGE
PLUMB, IN THE PLANE OF THE SUPPORT LINE

CONNECTION PLATE

PL 7" x 1/2"
IN THE CROSS-FRAME PLANE
NORMAL TO THE FLANGES

INTERMEDIATE STIFFENER

PL 6" x 1/2"
TIGHT FIT TO TOP FLANGE
MAX SPACING 12'-0", INTERIOR FACE

FLANGE-TO-WEB WELDS: 5/16" FILLET BOTH SIDES, CONTINUOUS. STIFFENER WELDS STOP AT THE CLIPS (DO NOT WRAP).

STIFFENER DETAILS

Sections looking along the girder; web broken (not to scale vertically)

STRUCTURAL STEEL NOTES

1. GIRDER PLATES, STIFFENERS AND GUSSETS: ASTM A709 GR. 50. CROSS-FRAME ANGLES AND CHANNELS: ASTM A709 GR. 36.
2. HIGH-STRENGTH BOLTS: ASTM F3125 GR. A325, 7/8" DIA, STANDARD HOLES, 1 1/2" MIN EDGE DISTANCE.
3. WELDING: AASHTO/AWS D1.5 BRIDGE WELDING CODE.
4. CROSS FRAMES: INTERMEDIATE FRAMES AND CONNECTION PLATES ARE NORMAL TO THE FLANGES; SUPPORT FRAMES AND BEARING STIFFENERS ARE PLUMB.
5. CROSS FRAMES PARALLEL TO SKEW IF SKEW <= 20 DEG, OTHERWISE NORMAL TO THE GIRDERS (AASHTO 6.7.4.2).
6. FIELD SPLICES: SLIP-CRITICAL BOLTED, NSBA DETAILS.
7. SHEAR STUDS: 7/8" DIA, 3 PER ROW @ 12".
8. CROSS-FRAME DETAILS SHOW THE FRAME AS DETAILED IN THE 3D MODEL; SEE THE IFC MODEL FOR EVERY FRAME.

SCALES

Intermediate cross frame: 1/4" = 1'-0"
Support cross frame: 1/4" = 1'-0"
Stiffener details: 3/4" = 1'-0"

GEOMETRY FROM THE PARAMETRIC MODEL. NOT FOR CONSTRUCTION UNLESS SEALED.

PROJECT

SAMPLE PROJECT

STEEL BRIDGE MODELER

STEEL DETAILS

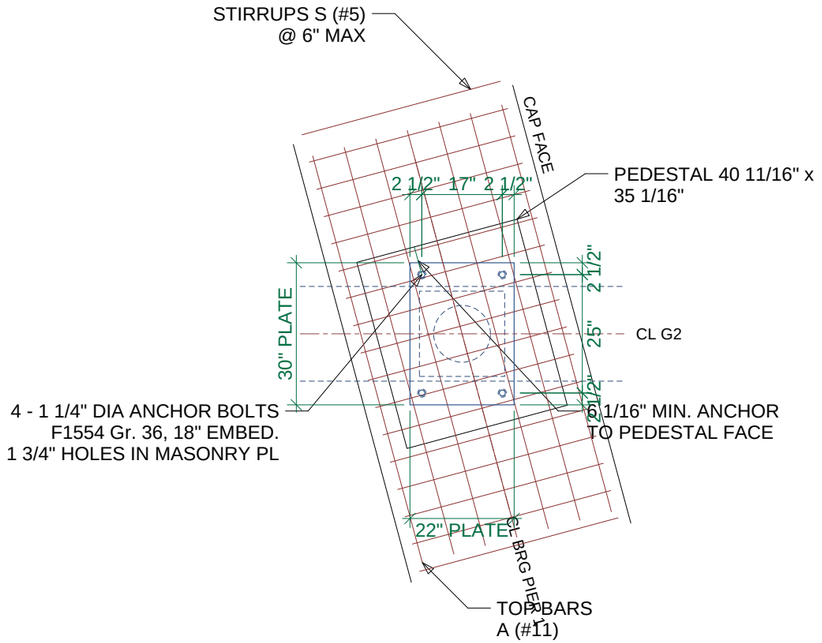
Cross frames and stiffeners

SAMPLE BRIDGE

SHEET

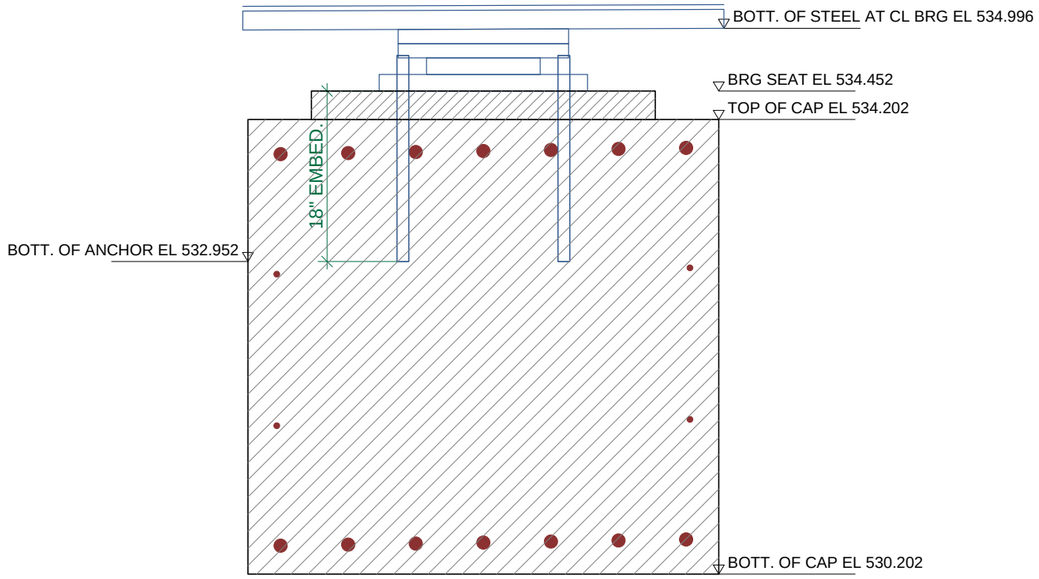
S-6 (6 OF 14)

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BEARING PLAN — B-Pier1-G2

Girder axis horizontal, up-station to the right; the bearing with the smallest anchor margin is shown



SECTION ALONG G2 AT PIER 1

Sole plate, upper bearing plate, disc, masonry plate (Fixed)

PIER 1 — ANCHOR BOLT CHECKS

BEARING	ANCHORS	EMBED.	EDGE TO PED.	EDGE TO CAP	BOTTOM EL	CAP BOTT. EL	CLEAR TO BARS	NEAREST BAR	STATUS
B-Pier1-G1	4 - 1 1/4"	18"	6 1/16"	12 1/2"	532.403	529.653	-7/16"	S (#5)	CHECK: pedestal edge, hits rebar
B-Pier1-G2	4 - 1 1/4"	18"	6 1/16"	12 9/16"	532.952	530.202	-1/2"	S (#5)	CHECK: pedestal edge, hits rebar
B-Pier1-G3	4 - 1 1/4"	18"	6 1/16"	12 9/16"	533.501	530.751	1/16"	V2 (#9)	CHECK: pedestal edge, rebar clearance
B-Pier1-G4	4 - 1 1/4"	18"	6 1/16"	12 9/16"	534.049	531.299	-3/8"	S (#5)	CHECK: pedestal edge, hits rebar
B-Pier1-G5	4 - 1 1/4"	18"	6 1/16"	12 9/16"	534.597	531.847	-7/16"	S (#5)	CHECK: pedestal edge, hits rebar

Edge distance target: max(6 x anchor dia, 4"); clear distance to reinforcing target: 1"; anchors must stop 3" above the bottom of cap.
Clearances are measured in 3D between each anchor bolt and the generated cap bars (stirrups, top, side and bottom bars).

SCALES
Bearing plan: 3/8" = 1'-0"
Section along girder: 3/4" = 1'-0"

GEOMETRY FROM THE PARAMETRIC MODEL. NOT FOR CONSTRUCTION UNLESS SEALED.

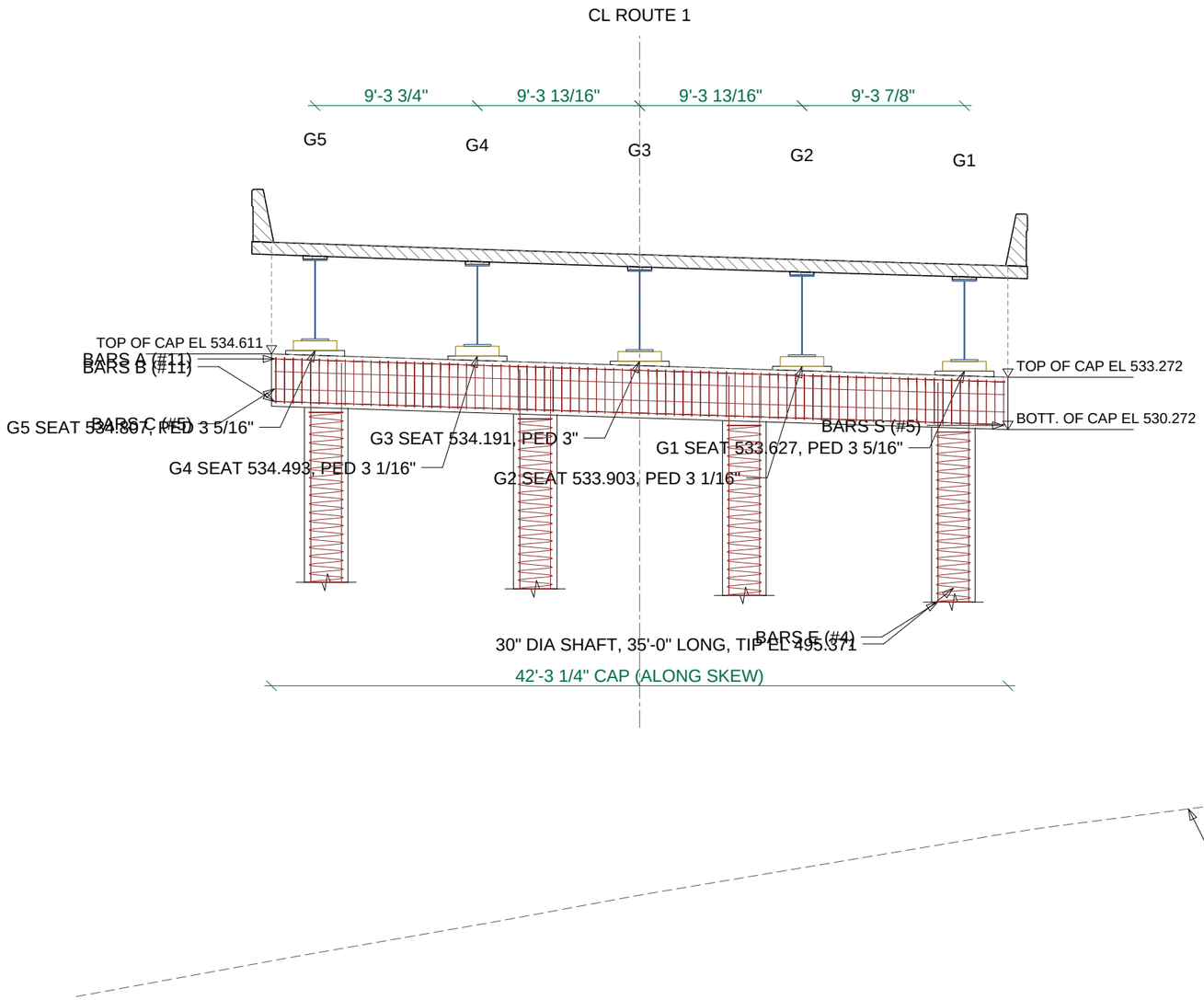
PROJECT
SAMPLE PROJECT
STEEL BRIDGE MODELER

PIER 1 BEARING DETAILS

Anchor bolts, pedestal and cap clearances

SAMPLE BRIDGE

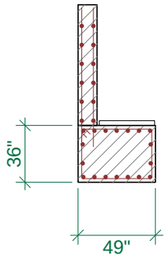
SHEET
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ABUT 1 — ABUTMENT ELEVATION

Looking down-station toward the backwall, dimensions along the support line (skew 15°00'00.00") · Scale 1/8" = 1'-0"

CAP TOP: SLOPED 3.167% (BEST FIT THROUGH THE SEATS)
PEDESTALS: 3" TO 3 5/16" HIGH AT CL BEARING

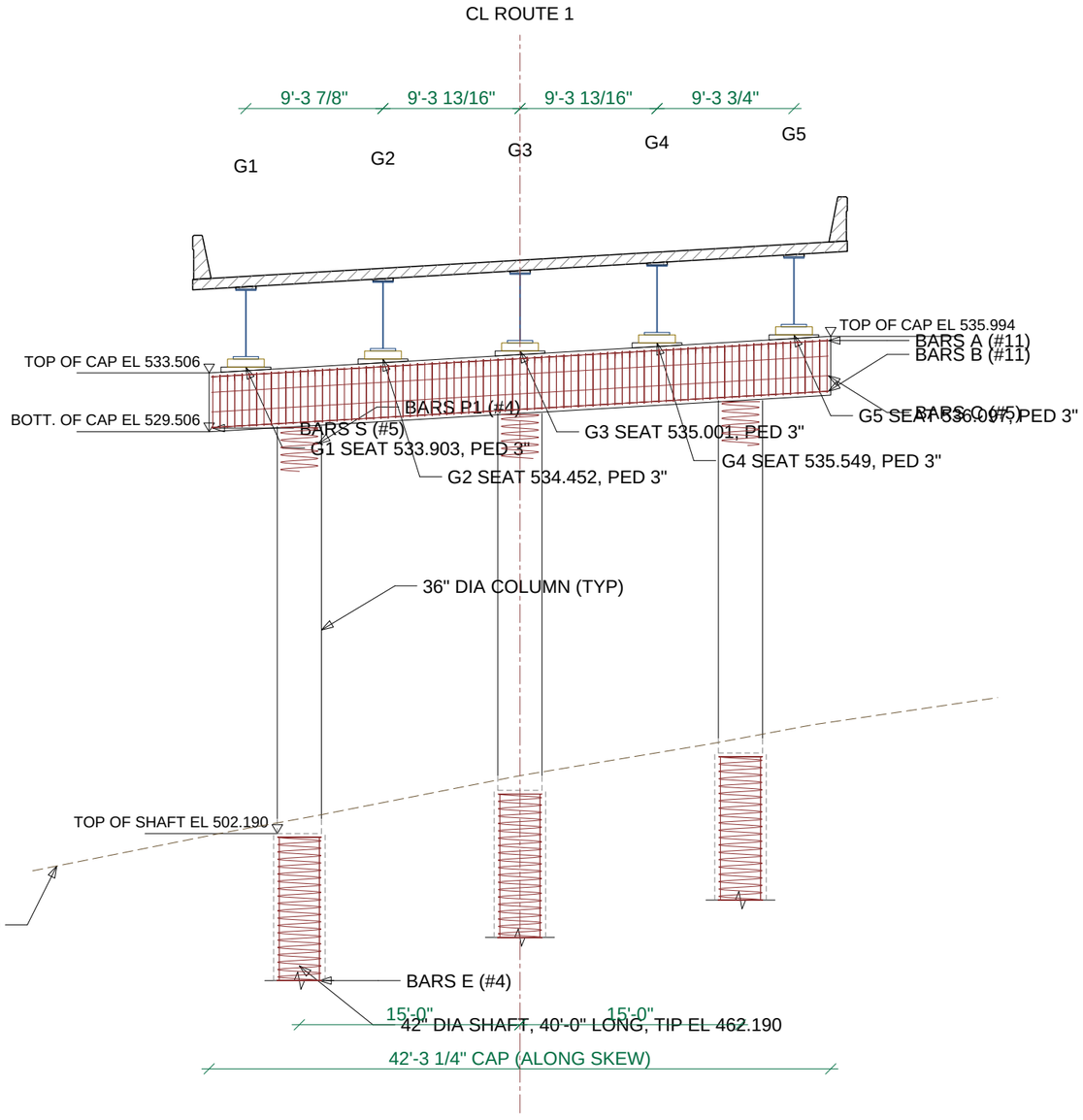


BARS A: 7 #11 TOP
BARS B: 7 #11 BOTTOM
BARS C: 2 #5 EACH FACE
BARS S: #5 STIRRUPS @ 6" MAX

SECTION A–A

Through the cap at the CL alignment · Scale 1/8" = 1'-0"

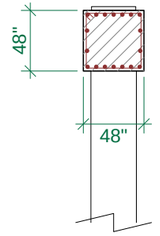
SCALES	PROJECT		SHEET
Abut 1: 1/8" = 1'-0"	SAMPLE PROJECT	ABUT 1 ABUTMENT Elevation and cap section SAMPLE BRIDGE	S-8 (8 OF 14)
GEOMETRY FROM THE PARAMETRIC MODEL. NOT FOR CONSTRUCTION UNLESS SEALED.	STEEL BRIDGE MODELER		10/08/2026



PIER 1 — BENT ELEVATION

Looking up-station, dimensions along the support line (skew 15°00'00.00") · Scale 1" = 10'-0"

CAP TOP: SLOPED 5.887% (BEST FIT THROUGH THE SEATS)
PEDESTALS: 3" TO 3" HIGH AT CL BEARING



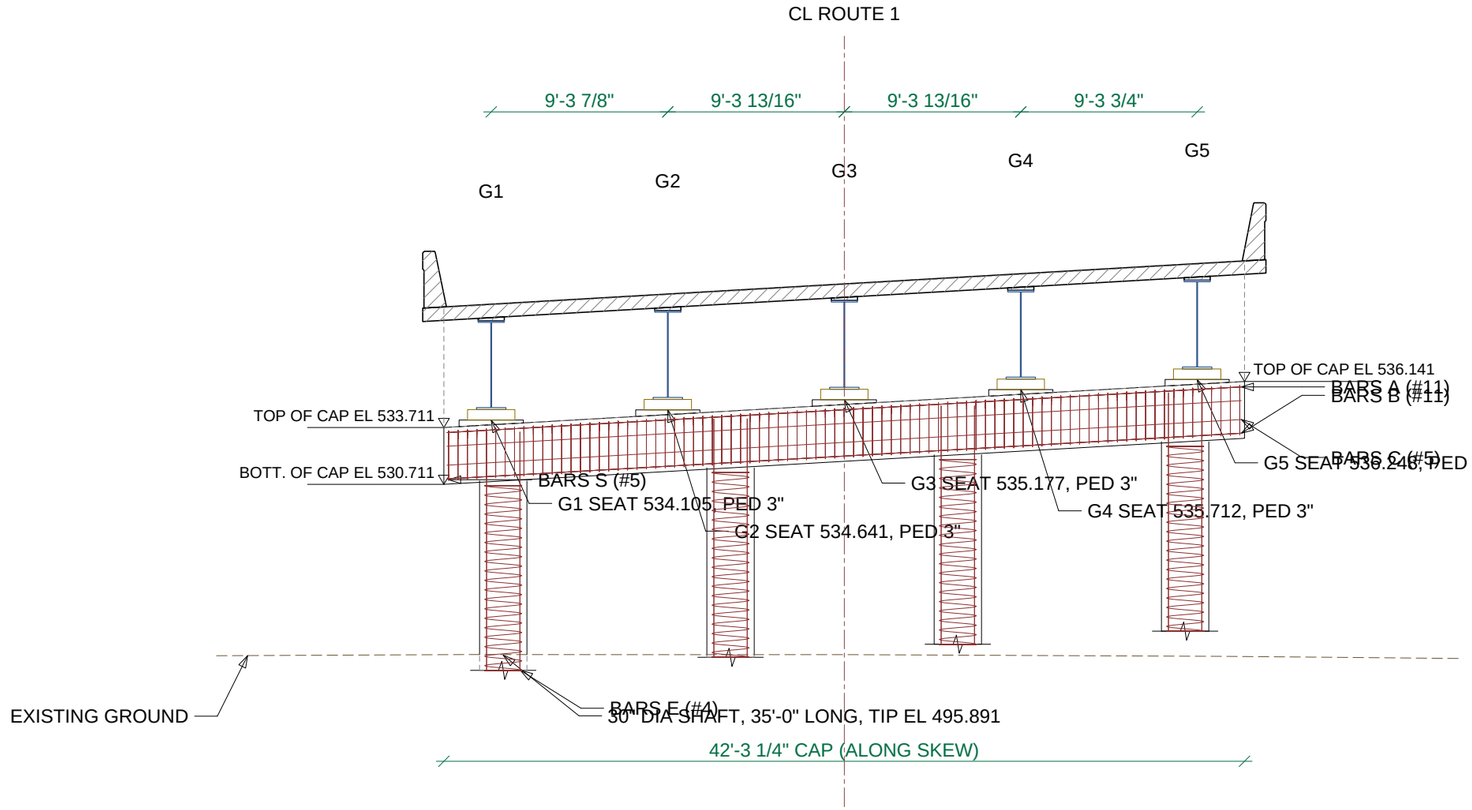
BARS A: 7 #11 TOP
BARS B: 7 #11 BOTTOM
BARS C: 2 #5 EACH FACE
BARS S: #5 STIRRUPS @ 6" MAX

SECTION A–A

Through the cap at the CL alignment · Scale 1" = 10'-0"

SCALES Pier 1: 1" = 10'-0"	PROJECT SAMPLE PROJECT STEEL BRIDGE MODELER	PIER 1 BENT Elevation and cap section SAMPLE BRIDGE	SHEET S-9 (9 OF 14) 10/08/2026

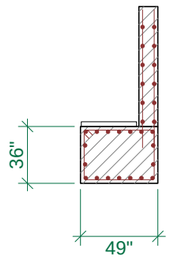
GEOMETRY FROM THE PARAMETRIC MODEL. NOT FOR CONSTRUCTION UNLESS SEALED.



ABUT 2 — ABUTMENT ELEVATION

Looking up-station toward the backwall, dimensions along the support line (skew 15°00'00.00") · Scale 1/8" = 1'-0"

CAP TOP: SLOPED 5.750% (BEST FIT THROUGH THE SEATS)
PEDESTALS: 3" TO 3" HIGH AT CL BEARING



BARS A: 7 #11 TOP
BARS B: 7 #11 BOTTOM
BARS C: 2 #5 EACH FACE
BARS S: #5 STIRRUPS @ 6" MAX

SECTION A–A

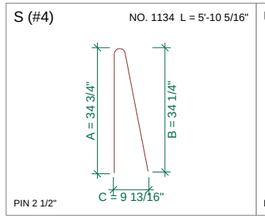
Through the cap at the CL alignment · Scale 1/8" = 1'-0"

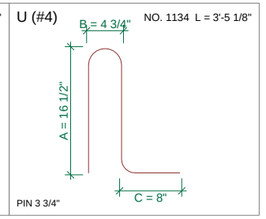
SCALES Abut 2: 1/8" = 1'-0"	PROJECT SAMPLE PROJECT STEEL BRIDGE MODELER	ABUT 2 ABUTMENT Elevation and cap section SAMPLE BRIDGE	SHEET S-10 (10 OF 14) 10/08/2026

GEOMETRY FROM THE PARAMETRIC MODEL. NOT FOR CONSTRUCTION UNLESS SEALED.

SUPERSTRUCTURE

MARK	SIZE	NO.	LENGTH	TYPE	A	B	C / D	WEIGHT	LOCATION
T1	#5	561	44'-3"	STRAIGHT				25,892	DECK TOP TRANSVERSE (EPOXY)
T2	#5	5	44'-2"	STRAIGHT				230	DECK TOP TRANSVERSE (EPOXY)
B1	#5	561	44'-3"	STRAIGHT				25,892	DECK BOTTOM TRANSVERSE (EPOXY)
B2	#5	5	44'-2"	STRAIGHT				230	DECK BOTTOM TRANSVERSE (EPOXY)
L1	#4	232	60'-0"	STRAIGHT				9,299	DECK TOP LONGITUDINAL (EPOXY)
L2	#4	29	30'-0"	STRAIGHT				581	DECK TOP LONGITUDINAL (EPOXY)
L3	#4	12	51'-2"–53'-11"	STRAIGHT				421	DECK TOP LONGITUDINAL (EPOXY)
L4	#4	12	48'-1"–50'-11"	STRAIGHT				397	DECK TOP LONGITUDINAL (EPOXY)
L5	#4	5	46'-10"–47'-10"	STRAIGHT				158	DECK TOP LONGITUDINAL (EPOXY)
L6	#4	12	23'-2"–26'-0"	STRAIGHT				197	DECK TOP LONGITUDINAL (EPOXY)
L7	#4	12	20'-1"–22'-11"	STRAIGHT				173	DECK TOP LONGITUDINAL (EPOXY)
L8	#4	5	18'-10"–19'-10"	STRAIGHT				65	DECK TOP LONGITUDINAL (EPOXY)
M1	#5	232	60'-0"	STRAIGHT				14,519	DECK BOTTOM LONGITUDINAL (EPOXY)
M2	#5	29	30'-0"	STRAIGHT				907	DECK BOTTOM LONGITUDINAL (EPOXY)
M3	#5	12	53'-2"–55'-11"	STRAIGHT				683	DECK BOTTOM LONGITUDINAL (EPOXY)
M4	#5	12	50'-1"–52'-11"	STRAIGHT				644	DECK BOTTOM LONGITUDINAL (EPOXY)
M5	#5	5	48'-10"–49'-10"	STRAIGHT				257	DECK BOTTOM LONGITUDINAL (EPOXY)
M6	#5	12	25'-8"–28'-6"	STRAIGHT				339	DECK BOTTOM LONGITUDINAL (EPOXY)
M7	#5	12	22'-7"–25'-5"	STRAIGHT				301	DECK BOTTOM LONGITUDINAL (EPOXY)
M8	#5	5	21'-4"–22'-4"	STRAIGHT				114	DECK BOTTOM LONGITUDINAL (EPOXY)
N	#5	114	36'-9"	STRAIGHT				4,370	DECK NEGATIVE MOMENT, PIER 1 (EPOX…
S	#4	1134	5'-10 5/16"	S (SSTR)	34 3/4"	34 1/4"	9 13/16"	4,438	LEFT RAIL, RIGHT RAIL (EPOXY)
U	#4	1134	3'-5 1/8"	U (SSTR)	16 1/2"	4 3/4"	8"	2,597	LEFT RAIL, RIGHT RAIL (EPOXY)
R1	#4	36	59'-1"	STRAIGHT				1,421	RIGHT RAIL (EPOXY)
R2	#4	4	58'-11"	STRAIGHT				157	RIGHT RAIL (EPOXY)
R3	#4	32	57'-10"	STRAIGHT				1,236	LEFT RAIL (EPOXY)
R4	#4	8	57'-6"	STRAIGHT				307	LEFT RAIL (EPOXY)





TOTAL REINFORCING 95,824 LB

SCALES

GEOMETRY FROM THE PARAMETRIC MODEL. NOT FOR CONSTRUCTION UNLESS SEALED.

PROJECT

SAMPLE PROJECT

STEEL BRIDGE MODELER

BAR LIST SUPERSTRUCTURE

Bar list and bending diagrams

SAMPLE BRIDGE

SHEET

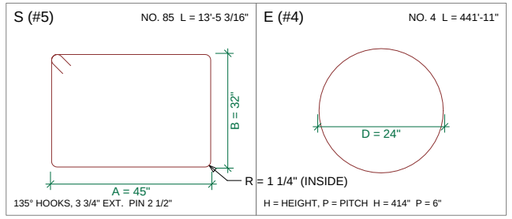
S-11 (11 OF 14)

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ABUT 1

MARK	SIZE	NO.	LENGTH	TYPE	A	B	C / D	WEIGHT	LOCATION
A	#11	7	41'-11"	STRAIGHT				1,559	ABUT 1 CAP TOP
B	#11	7	41'-11"	STRAIGHT				1,559	ABUT 1 CAP BOTTOM
C	#5	4	41'-11"	STRAIGHT				175	ABUT 1 CAP SIDE FACES
S	#5	85	13'-5 3/16"	TIE	45"	32"		1,191	ABUT 1 CAP STIRRUPS
W	#5	86	7'-4"	STRAIGHT				658	ABUT 1 BACKWALL VERTICALS
H	#5	12	41'-11"	STRAIGHT				525	ABUT 1 BACKWALL HORIZONTALS
D	#9	48	37'-4"	STRAIGHT				6,093	ABUT 1 DRILLED SHAFTS VERTICALS
E	#4	4	441'-11"	SPIRAL			24"	1,181	ABUT 1 DRILLED SHAFTS SPIRAL

TOTAL REINFORCING 12,940 LB

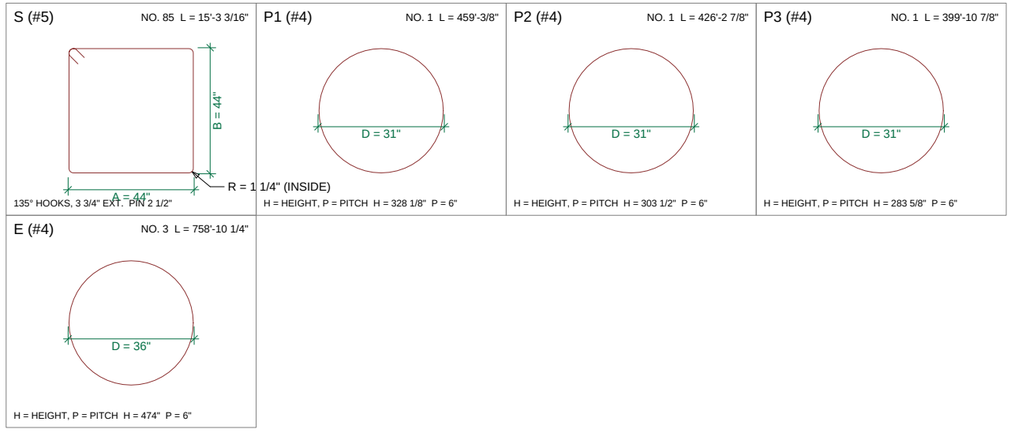


SCALES	PROJECT		SHEET
	SAMPLE PROJECT	BAR LIST ABUT 1 Bar list and bending diagrams SAMPLE BRIDGE	S-12 (12 OF 14)
GEOMETRY FROM THE PARAMETRIC MODEL. NOT FOR CONSTRUCTION UNLESS SEALED.	STEEL BRIDGE MODELER		10/08/2026

PIER 1

MARK	SIZE	NO.	LENGTH	TYPE	A	B	C / D	WEIGHT	LOCATION
A	#11	7	42'-0"	STRAIGHT				1,562	PIER 1 CAP TOP
B	#11	7	42'-0"	STRAIGHT				1,562	PIER 1 CAP BOTTOM
C	#5	4	42'-0"	STRAIGHT				175	PIER 1 CAP SIDE FACES
S	#5	85	15'-3 3/16"	TIE	44"	44"		1,353	PIER 1 CAP STIRRUPS
V1	#9	12	35'-0"	STRAIGHT				1,428	PIER 1 COLUMN 1 VERTICALS
V2	#9	12	32'-11"	STRAIGHT				1,343	PIER 1 COLUMN 2 VERTICALS
V3	#9	12	31'-4"	STRAIGHT				1,278	PIER 1 COLUMN 3 VERTICALS
P1	#4	1	459'-3/8"	SPIRAL			31"	307	PIER 1 COLUMN 1 SPIRAL
P2	#4	1	426'-2 7/8"	SPIRAL			31"	285	PIER 1 COLUMN 2 SPIRAL
P3	#4	1	399'-10 7/8"	SPIRAL			31"	267	PIER 1 COLUMN 3 SPIRAL
D	#9	36	39'-6"	STRAIGHT				4,835	PIER 1 DRILLED SHAFTS VERTICALS
E	#4	3	758'-10 1/4"	SPIRAL			36"	1,521	PIER 1 DRILLED SHAFTS SPIRAL

TOTAL REINFORCING 15,916 LB

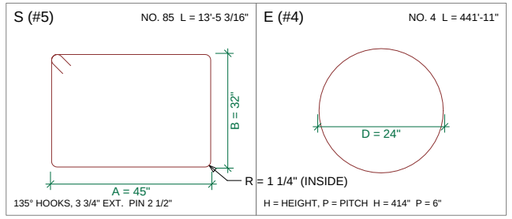


SCALES	PROJECT		SHEET
	SAMPLE PROJECT	BAR LIST PIER 1 Bar list and bending diagrams	S-13 (13 OF 14)
GEOMETRY FROM THE PARAMETRIC MODEL. NOT FOR CONSTRUCTION UNLESS SEALED.	STEEL BRIDGE MODELER	SAMPLE BRIDGE	10/08/2026

ABUT 2

MARK	SIZE	NO.	LENGTH	TYPE	A	B	C / D	WEIGHT	LOCATION
A	#11	7	42'-0"	STRAIGHT				1,562	ABUT 2 CAP TOP
B	#11	7	42'-0"	STRAIGHT				1,562	ABUT 2 CAP BOTTOM
C	#5	4	42'-0"	STRAIGHT				175	ABUT 2 CAP SIDE FACES
S	#5	85	13'-5 3/16"	TIE	45"	32"		1,191	ABUT 2 CAP STIRRUPS
W	#5	86	7'-3"	STRAIGHT				650	ABUT 2 BACKWALL VERTICALS
H	#5	12	42'-0"	STRAIGHT				526	ABUT 2 BACKWALL HORIZONTALS
D	#9	48	37'-4"	STRAIGHT				6,093	ABUT 2 DRILLED SHAFTS VERTICALS
E	#4	4	441'-11"	SPIRAL			24"	1,181	ABUT 2 DRILLED SHAFTS SPIRAL

TOTAL REINFORCING 12,940 LB



SCALES	PROJECT	BAR LIST ABUT 2	SHEET
	SAMPLE PROJECT	Bar list and bending diagrams	S-14 (14 OF 14)
GEOMETRY FROM THE PARAMETRIC MODEL. NOT FOR CONSTRUCTION UNLESS SEALED.	STEEL BRIDGE MODELER	SAMPLE BRIDGE	10/08/2026